MCQ solutions

1. **B.** The denominator tends to 0 through negative values. The side of approach determines the sign.
2. **C.** The square root is positive or zero; $\sqrt{x^2} = |x| = -x$ for $x < 0$. This is not an equation in two solutions.
3. **D.** Divide by x^2 gives $(3 - 1/x)/(1 + 4/x^2) \to 3$.
4. **A.** For $x \neq 1$, the quotient is $x + 1$. A forbidden value does not make a limit impossible.
5. **C.** The difference tends to 0 and is zero at $k\pi$, $k \geq 1$, with a change of sign. An asymptote can be crossed.
6. **B.** This is x^2/e^x, which tends to 0 by comparison of growth rates. The exponential dominates every power fixed.
7. **A.** Let $t = -x \to +\infty$: the expression is $-t/e^t \to 0$.
8. **D.** Multiplying by the conjugate gives 2/ $(\sqrt{1 + 2/x} + 1) \to 1$.
9. **C.** With $g = x$, take $f = 1/x^2$, $1/x$ or 1/ $\sqrt{x}$: we obtain respectively 0, 1 and $+\infty$.
10. **B.** The comparison theorem is enough for the limit. 2 x^2 disproves both limiting equalities; the inequality does not give monotonicity.

True/false. False: question 5. False: question 4. False: $x/x \to 1$, $x/x^2 \to 0$, $x^2/x \to +\infty$.

Solutions to the applications and guidance

Application 1. At -1^- : $+\infty$; at -1^+ : $-\infty$; at both infinities: 3. Asymptotes $x = -1$ and $y = 3$.

Application 2. Limits: 2; $\frac{4x}{\sqrt{x^2+4x}+x} = \frac{4}{\sqrt{1+4/x}+1} \to 2$; $\frac{1}{\sqrt{1+1/x}+1} \to 1/2$. The expressions are defined for x big enough.

Application 3. For example $x^2/x \to 0$, $x/x \to 1$, $x/x^2 = 1/x \to +\infty$ at 0^+. The functions are considered outside 0.

Application 4. The first tends to 0 by linearity and comparative growth. For the second, $t = -x \to +\infty$ gives $xe^x = -te^{-t} \to 0$. The symbol ∞ is not a real number at which we evaluate a function.

Application 5. $q(0) = 1$. $q(x) - 1$ is negative for $x < 0$, positive for $x > 0$. The limits concern the infinite ends of the domain, so they do not contradict this intersection.

Application 6. The limit to infinity is a, therefore $a = 2$; $h(0) = -b = 3$, so $b = -3$. Thus $h(x) = 2 - 1/(x-1)$: limits $+\infty$ at 1^-, $-\infty$ at 1^+ ; graph above 2 to the left of 1 and at bottom right.

Reading then calculation

Before any table, have students anticipate the sign of the quotient. A numerical result contradicting the sign expected should trigger a check. Have students draw p and q in the same window to dispute the image of an "impassable barrier".

Teaching notes and solutions

Guidance for the teacher

In S21, link each limit to a reading of the graph. In S23, require the signs of the denominator on the left and right. In S24, have students choose the transformation before calculating: factorisation, conjugate, local expansion not used. In S26, separate the proof of comparative growth compared in their use. Oblique asymptotes are not required here.

Solutions to the activity

1. Since $3 \quad x-2=3(x-1)+1$, for $\quad x \neq 1, f(x)=3+\frac{1}{x-1}$.

2. At $\pm\infty, \frac{1}{x-1} \to 0$: the limits equal 3. At $1^{-}, x-1 \to 0^{-}$, therefore $f(x) \to -\infty$; at $1^{+}, x-1 \to 0^{+}$, so $\quad f(x) \to +\infty$.

3. The sign of $\quad f(x)-3=\frac{1}{x-1}$ is negative for $\quad x<1$, positive for $\quad x>1$. The curve is therefore respectively below and above the asymptote $\quad y=3$.

4. For $\quad x>1, |f(x)-3|<0.01$ is equivalent to $\frac{1}{x-1}<0.01$, then $\quad x>101$. We can choose $\quad A=101$.

5. No: $\quad f(x)-3=\frac{1}{x-1}$ is never zero. The curve may appear to merge with the asymptote at the precision of the screen, but it does not meet it.

E1. For $\quad x \neq 0$, the quotient equals $\quad \frac{5-\frac{1}{x^2}}{2+\frac{7}{x^3}}$. In $\quad +\infty$, the reciprocal powers tend to 0 and the denominator to 2 $\quad \neq 0$. The limit is $\quad \frac{5}{2}$.

E2. At 2, $\quad (x-2)^2 \to 0$ through strictly positive values on both sides; its reciprocal therefore tends to $+\infty$. In $\quad \pm\infty$, the square tends to $\quad +\infty$ and its reciprocal to 0.

E3. For $\quad x \neq 1, x^2-1=(x-1)(x+1)$, so the quotient equals $\quad x+1$. Its limit at 1 is therefore 2. This simplification computes the limit; it does not make the original function defined at 1.

E4. Let $\quad t=2x \to +\infty$. Then $\quad x^3e^{-2x}=\frac{t^3}{8e^t} \to 0$, by comparative growth of the exponential and the power.

E5. The conjugate gives $\quad \sqrt{x^2+1}+x=\frac{1}{\sqrt{x^2+1}-x}$. When $\quad x \to -\infty$, both terms $\quad \sqrt{x^2+1}$ and $-x$ tend to $\quad +\infty$; the denominator therefore tends to $\quad +\infty$. The limit is 0, approached from above.

Solutions to homework DM3

1. Since $\quad x^2+1>0$ on $\quad \mathbb{R}$, the domain is $\quad \mathbb{R}$. By dividing by $\quad x^2$ for $\quad x \neq 0, f_a(x)=\frac{a+\frac{1}{x^2}}{1+\frac{1}{x^2}} \to a$ in $\pm\infty$.

2. $f_a(x)-a=\frac{ax^2+1-a(x^2+1)}{x^2+1}=\frac{1-a}{x^2+1}$. For $\quad a<1$, the curve is above $\quad y=a$; for $\quad a>1$, it is below; for $\quad a=1$, it coincides with this line.

3. By the quotient rule, $\quad f_a'(x)=\frac{[2ax(x^2+1)-2x(ax^2+1)]}{(x^2+1)^2}=\frac{2x(a-1}{(x^2+1)^2}$. If $\quad a>1$, the sign is that of $\quad x$: decreasing on $\quad]-\infty, 0]$, increasing on $\quad [0,+\infty[$; minimum $\quad f_a(0)=1$ and no maximum (the bound $\quad a$ is not reached). If $\quad a<1$, the variations are reversed: maximum 1 and no minimum. For $\quad a=1$, the function is constant, with value 1.

4. For $\quad a=3$ and $x \geq 0$, the inequality becomes $\quad \frac{2}{x^2+1} \leq 0{,}02$. The denominator is positive, so it is equivalent to to $x^2+1 \geq 100$, i.e. $\quad x \geq \quad \overline{99}$.

Solutions to the textbook or past-paper work

Exercise 26. When $x \to +\infty$, the functions $f(x) = x$ and $g(x) = x$ both tend to $+\infty$, but their difference is zero. With $f(x) = 2x$, $g(x) = x$, the difference tends to $+\infty$; by swapping f and g, it tends towards $-\infty$. These examples show that the difference is an indeterminate form.

Exercise 29. The denominator is zero at $\frac{3}{2}$, therefore $D_f = \mathbb{R} \setminus \{\frac{3}{2}\}$. We write

$$f(x) = \frac{5 - 2x}{2x - 3} = -1 + \frac{2}{2x - 3}.$$

Thus $f(x) \to -1$ at both infinities. At $\frac{3}{2}$, the numerator of the second term is positive and its denominator tends to 0: the left-hand limit is $-\infty$, the one on the right is $+\infty$. The asymptotes have equations $y = -1$ and $x = \frac{3}{2}$.